Often we create database queries dynamically, based on user input. A stray character or invalid user input can cause the query to return an error. Sometimes we will print the query itself. We also print the results of the error functions, such as mysql_error. The same applies to code unrelated to databases. Printing diagnostic information, even if it is as simple as saying "got here," can help.

9.9.5 Remote Debugging

You may enable remote debugging for all scripts by editing the configuration file php. ini, or you may use the debugger_on function. Once enabled, PHP will attempt to connect to a remote host and port each time a script is run.

You will need a port listening program to get the debugging information. There are numerous free port listeners for Windows. We have found Port Listener by Hauke X to work well. Similar programs exist for UNIX as well.

All debugging messages are sent to the listening host, regardless of the error reporting level set in php.ini. Messages are sent in a special format followed by a linefeed.

date time host(pid) type : message-data

The date is in the format YYYY-MM-DD. Time is in the format HH:MM:UUUUUU. The last six digits of time are the seconds and microseconds. The host is the name of the server and pid is the process identifier. The type is a special code described in Table 3. The rest of the line is a message terminated with a linefeed.

Using the remote debugger is as simple as executing a PHP script and watching the debug information appear in the port listener output.

Table 3. Debugging types

Type	Description
end	The end of an error message
frames	The number of frames in the call stack.
function	The name of the function where the error occurred.
location	The filename and line number that generated the error message.
message	A PHP error message.
start	This type signifies the beginning of a debugging message. The data-message for this line will be the type of error.